

Report No.:

Test Time: 2024/10/15 11:25:44

LUMINAIRE PHOTOMETRIC TEST REPORT

Luminaire Property

Manufacturer: 0
Luminaire Type: AL9332 4000K 80RA
Lamp Type:
Luminous Width: 0 mm
Luminous Length: 0 mm
Luminous Height: 0 mm

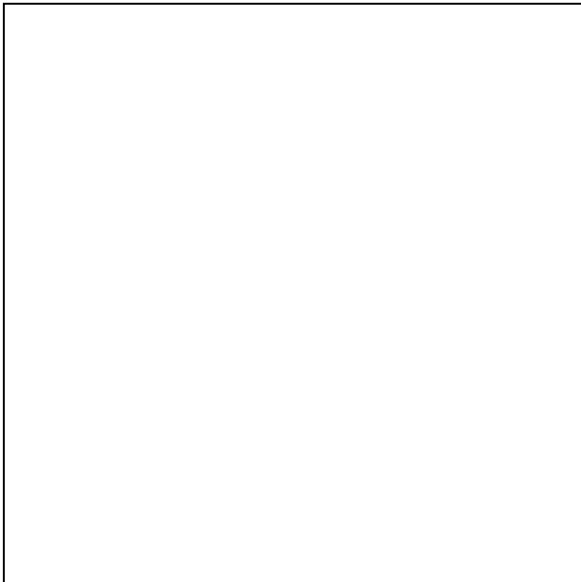
Voltage(V): 219.40 V
Current(A): 0.1050 A
Power(W): 12.83 W
Power Factor: 0.5540
Lamp Luminous Flux: 1000.92

Photometric Results

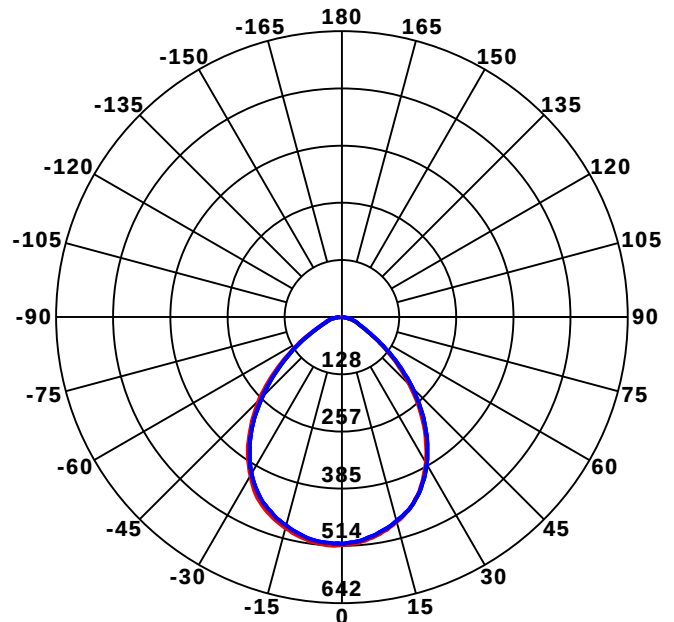
CIE Class: Direct
Luminaire Luminous Flux: 1000.922 lm
Downward Ratio: 100.00%
Luminaire Efficacy Rating(LER): 78.01 lm/W
Beam Angle(50%) C0-180,C90-270: 86.9,87.1
Field Angle(10%) C0-180,C90-270: 132.2,131.8
S/MH C0-180,C90-270: 1.155,1.155

Max. Intensity: 513.91 cd
Max.Intensity Angle: C=180 Gamma=2
Upward Ratio: 0.00%
Efficacy: 100.00%
Center Intensity: 512.54 cd
Effective Luminous Flux: 943.89 lm
ERP(ϕ use)(120deg): 901.59 lm

Picture of Luminaire



Luminous Intensity Distribution Curve



Beam Angle(50%) C0-180,C90-270: 86.9 deg,87.1 deg

C0-180 — C90-270 —

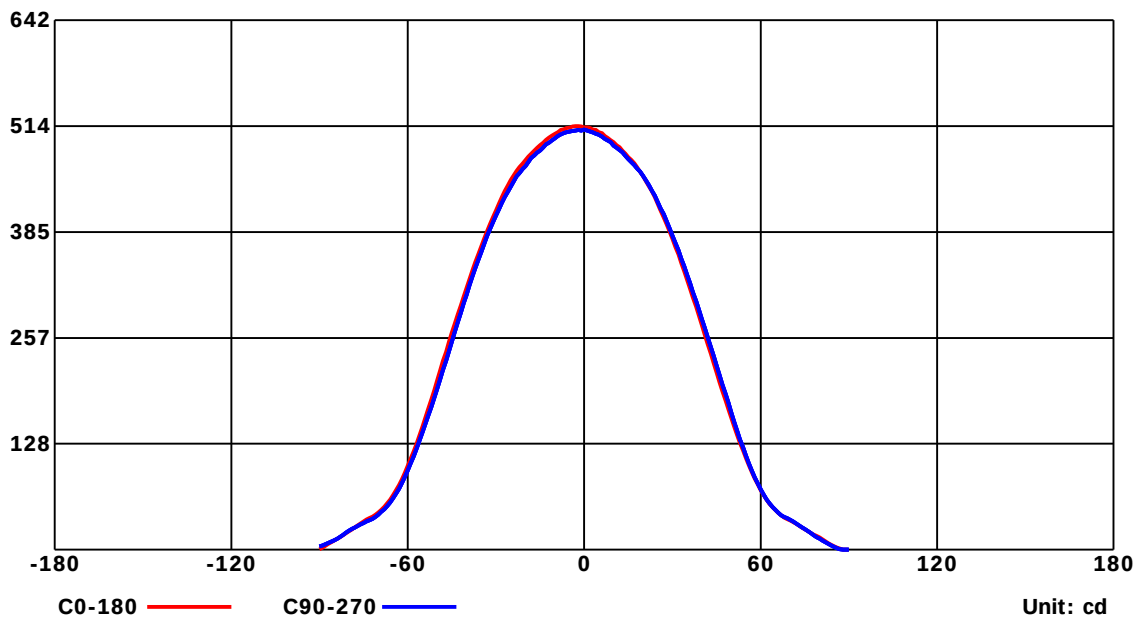
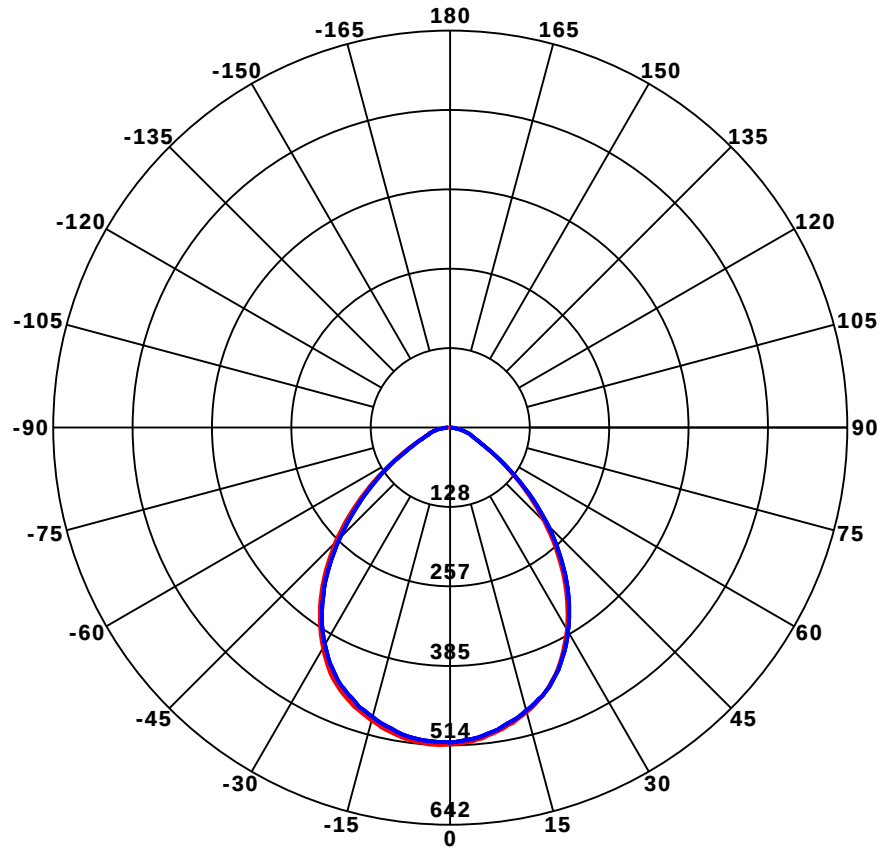
C Range: 0 - 360 Deg
C Interval: 45.0 Deg
Test Speed: High
Temperature: 6.25
Operators:
Test Date: 2024/10/15 11:25:44

G Range: 0 - 90 Deg
G Interval: 1.0 Deg
Test System: Huipu CPM-2000B Goniophotometer
Humidity: 1
Test Distance: 6.25 m
Remarks:

Report No.:

Test Time: 2024/10/15 11:25:44

Luminous Intensity Distribution Curve



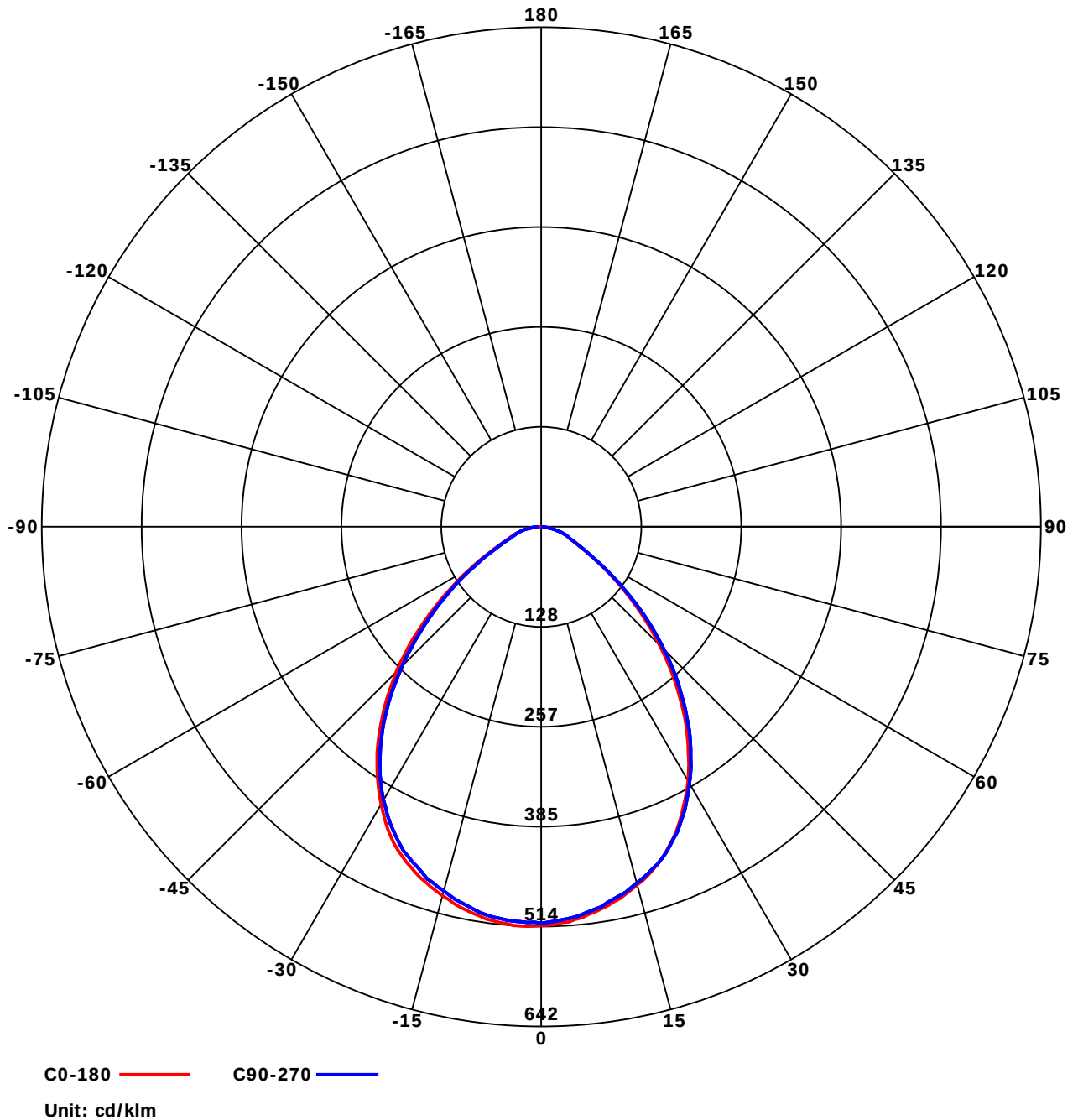
C Range: 0 - 360 Deg
C Interval: 45.0 Deg
Test Speed: High
Temperature: 6.25
Operators:
Test Date: 2024/10/15 11:25:44

G Range: 0 - 90 Deg
G Interval: 1.0 Deg
Test System: Huipu CPM-2000B Goniophotometer
Humidity: 1
Test Distance: 6.25 m
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Luminous Intensity Distribution Curve



C Range: 0 - 360 Deg
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Operators:
Test Date: 2024/10/15 11:25:44

G Range: 0 - 90 Deg
G Interval: 1.0 Deg
Test System: Huipu CPM-2000B Goniophotometer
Humidity: 1
Test Distance: 6.25 m
Remarks:

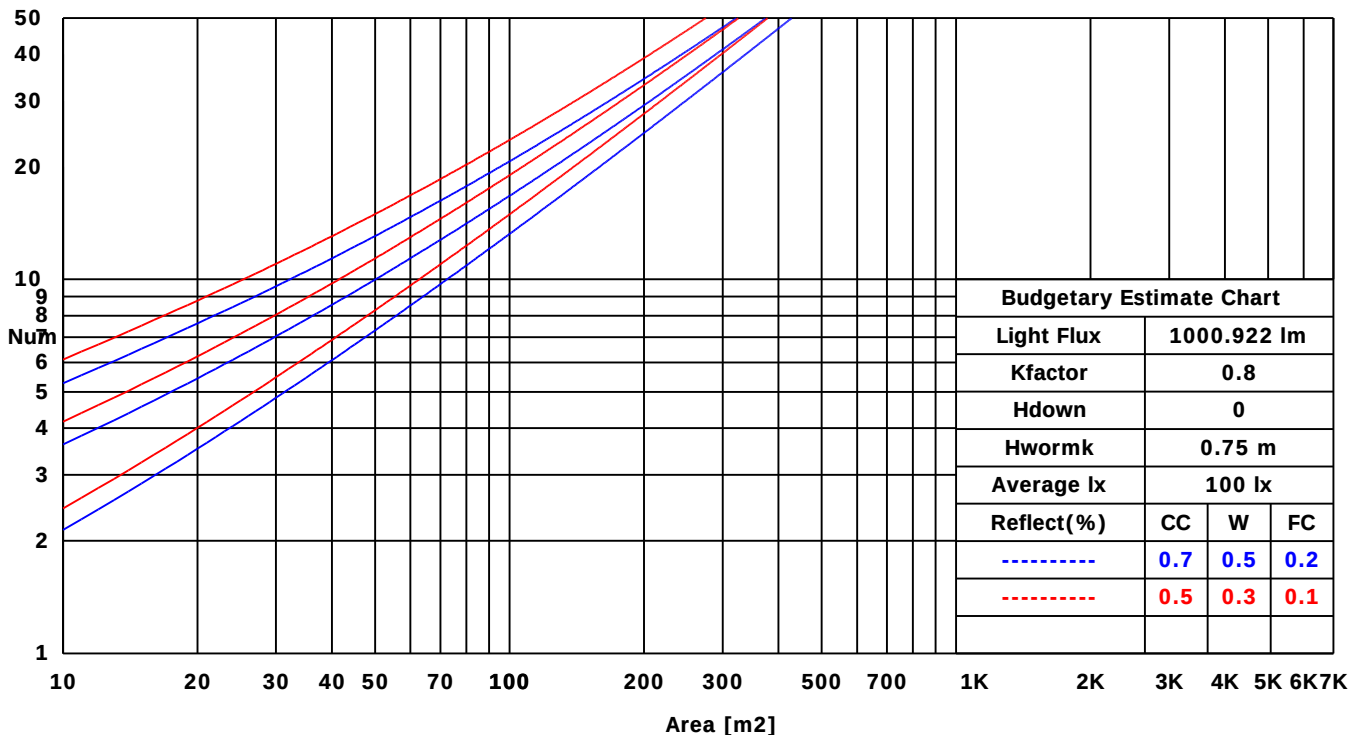
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UTILIZATION FACTORS TABLE

RHOCC	80			70			50			30			10			0
RHOW	50	30	10	50	30	10	50	30	10	50	30	10	50	30	10	0
RCR	COEFFICIENTS OF UTILIZATION FOR RHOFC=20															
0	1.18	1.18	1.18	1.15	1.15	1.15	1.10	1.10	1.10	1.05	1.05	1.05	1.01	1.01	1.01	0.99
1	1.08	1.07	1.06	1.07	1.05	1.04	1.03	1.01	1.00	0.98	0.96	0.94	0.91	0.89	0.87	0.82
2	0.95	0.93	0.92	0.94	0.92	0.90	0.92	0.89	0.86	0.88	0.85	0.82	0.83	0.79	0.76	0.72
3	0.84	0.82	0.80	0.84	0.81	0.79	0.82	0.78	0.75	0.79	0.75	0.72	0.75	0.71	0.67	0.63
4	0.75	0.72	0.71	0.75	0.72	0.69	0.73	0.69	0.66	0.71	0.67	0.63	0.69	0.64	0.59	0.55
5	0.67	0.64	0.63	0.67	0.64	0.62	0.66	0.62	0.59	0.65	0.60	0.56	0.63	0.57	0.53	0.49
6	0.60	0.58	0.56	0.60	0.57	0.55	0.60	0.56	0.53	0.59	0.54	0.50	0.58	0.52	0.47	0.44
7	0.54	0.52	0.51	0.55	0.52	0.50	0.55	0.51	0.47	0.54	0.49	0.45	0.54	0.47	0.43	0.39
8	0.49	0.47	0.46	0.50	0.47	0.45	0.50	0.46	0.43	0.50	0.45	0.41	0.50	0.44	0.39	0.36
9	0.45	0.43	0.42	0.46	0.43	0.41	0.46	0.42	0.39	0.46	0.41	0.37	0.46	0.40	0.36	0.33
10	0.42	0.39	0.38	0.42	0.39	0.37	0.43	0.39	0.36	0.43	0.38	0.34	0.43	0.37	0.33	0.30

LUMINAIRE BUDGETARY ESTIMATE DIAGRAM



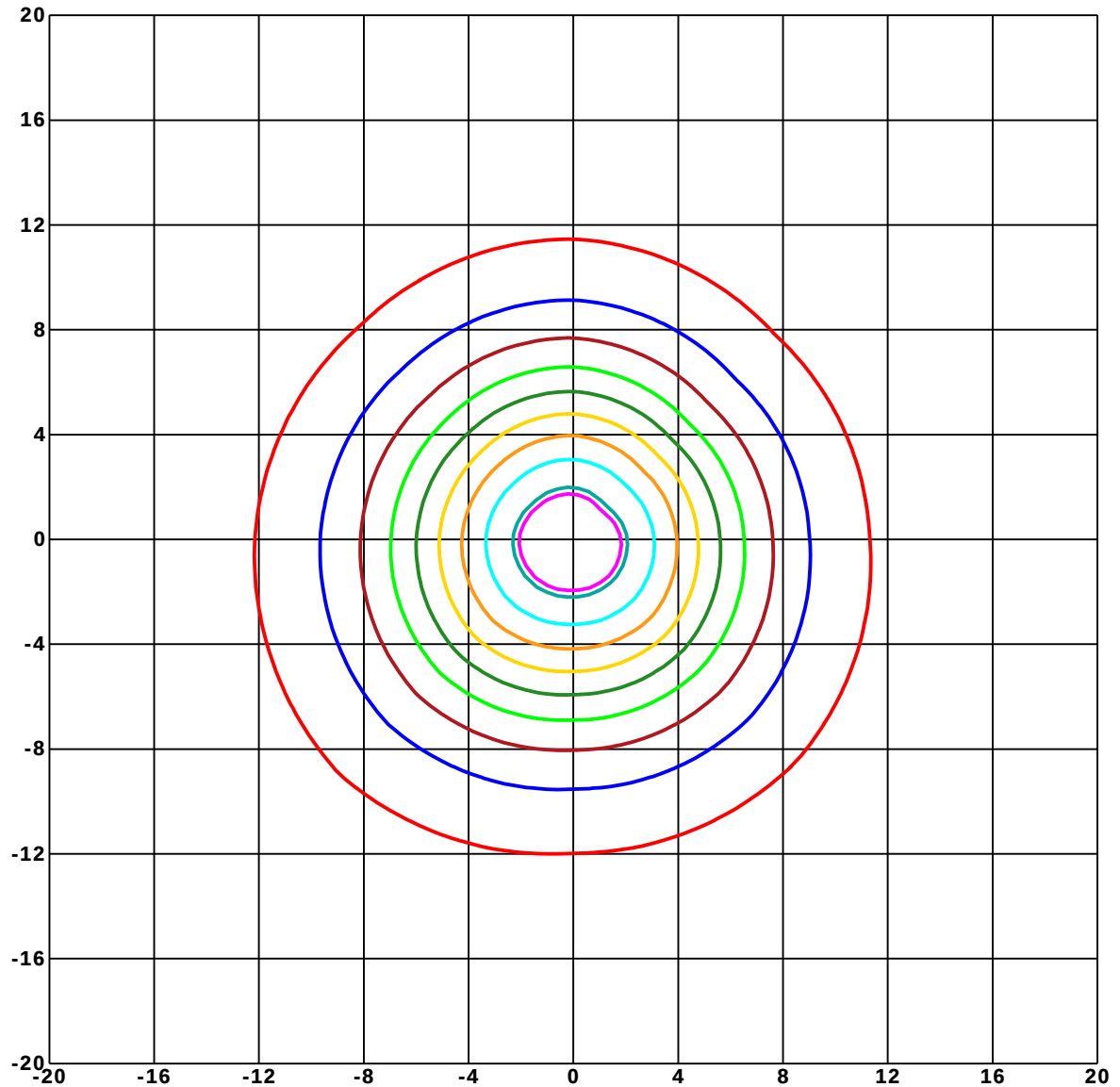
C Range: 0 - 360 Deg
 C Interval: 45.0 Deg
 Test Speed: High
 Temperature: 6.25
 Operators:
 Test Date: 2024/10/15 11:25:44

G Range: 0 - 90 Deg
 G Interval: 1.0 Deg
 Test System: Huipu CPM-2000B Goniophotometer
 Humidity: 1
 Test Distance: 6.25 m
 Remarks:

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ISOLX DIAGRAM



Unit of illumination:lx Height of plane:10.0(m)/32.81(ft)

4.721(92%Emax)	2.566(50%Emax)
4.618(90%Emax)	2.053(40%Emax)
4.105(80%Emax)	1.539(30%Emax)
3.592(70%Emax)	1.026(20%Emax)
3.079(60%Emax)	0.513(10%Emax)

C Range: 0 - 360 Deg

C Interval: 45.0 Deg

Test Speed: High

Temperature: 6.25

Operators:

Test Date: 2024/10/15 11:25:44

G Range: 0 - 90 Deg

G Interval: 1.0 Deg

Test System: Huipu CPM-2000B Goniophotometer

Humidity: 1

Test Distance: 6.25 m

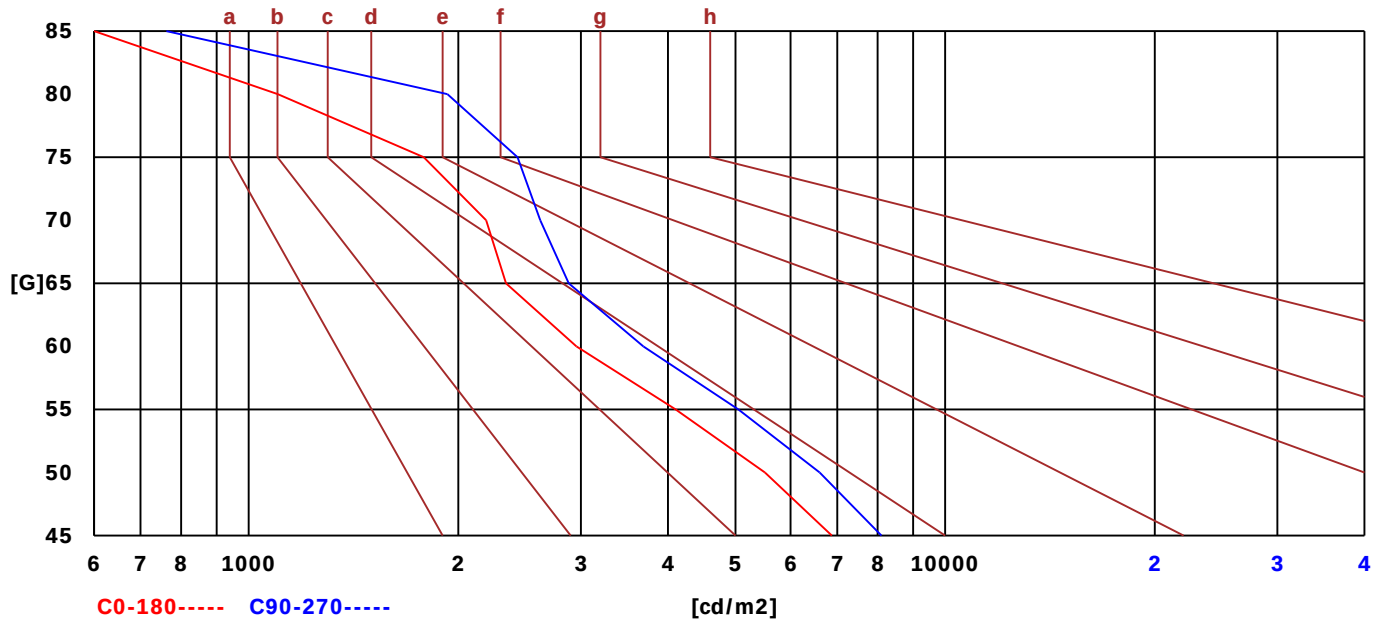
Remarks:

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BRIGHT LIMIT DIAGRAM

C/G	45	50	55	60	65	70	75	80	85
C0	6880	5512	4102	2957	2343	2193	1785	1102	0
C90	8103	6609	5048	3689	2881	2624	2434	1929	762
Dazzle	Quality	Illuminance	---	---	---	---	---	---	---
1.15	A	2000	1000	500	<=300				
1.5	B		2000	1000	500	<=300			
1.85	C			2000	1000	500	<=300		
2.2	D				2000	1000	500	<=300	
2.55	E					2000	1000	500	<=300
---	---	a	b	c	d	e	f	g	h



C Range: 0 - 360 Deg

C Interval: 45.0 Deg

Test Speed: High

Temperature: 6.25

Operators:

Test Date: 2024/10/15 11:25:44

G Range: 0 - 90 Deg

G Interval: 1.0 Deg

Test System: Huipu CPM-2000B Goniophotometer

Humidity: 1

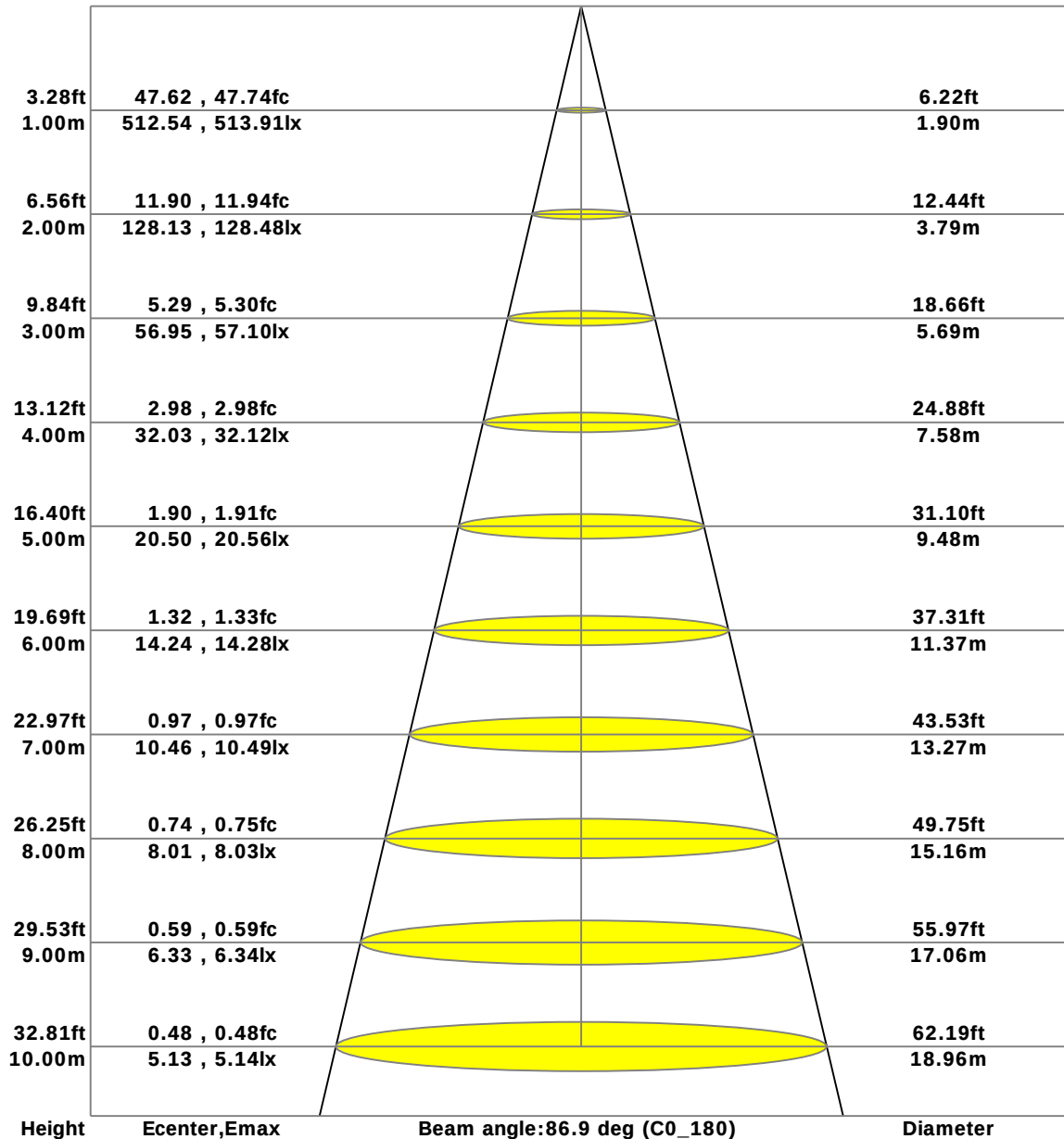
Test Distance: 6.25 m

Remarks:

Report No.:

Test Time: 2024/10/15 11:25:44

LX-DISTANCE DIAGRAM



C Range: 0 - 360 Deg

C Interval: 45.0 Deg

Test Speed: High

Temperature: 6.25

Operators:

Test Date: 2024/10/15 11:25:44

G Range: 0 - 90 Deg

G Interval: 1.0 Deg

Test System: Huipu CPM-2000B Goniophotometer

Humidity: 1

Test Distance: 6.25 m

Remarks:

Report No.:

Test Time: 2024/10/15 11:25:44

UGR(Unified Glare Rating) TABLE

RHOCC	70	70	50	50	30	70	70	50	50	30
RHOW	50	30	50	30	30	50	30	50	30	30
RHOF	20	20	20	20	20	20	20	20	20	20
Room Size X Y	Viewed Endwise					Viewed Crosswise				
2H 2H	17.11	17.68	17.03	17.87	18.11	17.19	17.76	17.11	17.94	18.19
3H	16.78	17.31	16.77	17.53	17.82	16.86	17.39	16.85	17.61	17.90
4H	16.60	17.10	16.62	17.33	17.65	16.67	17.17	16.70	17.41	17.73
6H	16.39	16.87	16.45	17.12	17.47	16.47	16.94	16.53	17.20	17.55
8H	16.29	16.74	16.37	17.01	17.38	16.36	16.82	16.45	17.09	17.45
12H	16.18	16.62	16.28	16.90	17.28	16.26	16.70	16.36	16.97	17.35
4H 2H	17.65	18.15	17.68	18.39	18.71	17.73	18.23	17.75	18.46	18.78
3H	17.38	17.82	17.48	18.09	18.47	17.45	17.89	17.55	18.16	18.55
4H	17.19	17.60	17.34	17.90	18.31	17.26	17.67	17.41	17.97	18.38
6H	16.99	17.35	17.18	17.67	18.12	17.06	17.42	17.25	17.74	18.19
8H	16.88	17.21	17.10	17.55	18.02	16.95	17.28	17.17	17.62	18.09
12H	16.77	17.07	17.01	17.42	17.92	16.84	17.14	17.08	17.50	17.99
8H 4H	17.68	18.13	17.76	18.40	18.76	17.71	18.16	17.79	18.43	18.79
6H	17.07	17.35	17.34	17.71	18.23	17.09	17.37	17.36	17.73	18.25
8H	16.96	17.21	17.26	17.59	18.12	16.98	17.23	17.28	17.61	18.14
12H	16.85	17.07	17.18	17.46	18.01	16.87	17.09	17.19	17.48	18.03
12H 4H	17.20	17.51	17.45	17.86	18.36	17.21	17.52	17.46	17.87	18.36
6H	17.00	17.25	17.30	17.63	18.17	17.01	17.26	17.31	17.64	18.17

C Range: 0 - 360 Deg
 C Interval: 45.0 Deg
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 Operators:
 Test Date: 2024/10/15 11:25:44

G Range: 0 - 90 Deg
 G Interval: 1.0 Deg
 Test System: Huipu CPM-2000B Goniophotometer
 Humidity: 1
 Test Distance: 6.25 m
 Remarks:

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ZONAL LUMINOUS FLUX TABLE-1

[illegible]

C Range: 0 - 360 Deg
C Interval: 45.0 Deg
Test Speed: High
Temperature: 6.25
Operators:
Test Date: 2024/10/15 11:25:44

G Range: 0 - 90 Deg
G Interval: 1.0 Deg
Test System: Huipu CPM-2000B Goniophotometer
Humidity: 1
Test Distance: 6.25 m
Remarks:

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ZONAL LUMINOUS FLUX TABLE-2

γ	C0	C45	C90	C135	C180	C225	C270	C315	γ	φ zonal	φ total	%lum	%lamp
10	494.5	481.4	490.4	495.5	504.6	503.0	499.2	500.0	10	48.0	48.0	4.8	4.8
20	453.4	437.4	453.6	457.9	472.9	476.0	465.4	467.7	20	135.5	183.5	18.3	18.3
30	377.3	354.4	381.1	387.3	411.9	428.6	405.5	408.8	30	198.4	381.9	38.2	38.2
40	272.2	248.2	279.7	288.0	317.7	342.6	309.4	313.8	40	217.0	598.9	59.8	59.8
50	159.7	141.7	166.4	169.9	205.4	232.3	195.2	197.1	50	184.7	783.5	78.3	78.3
60	72.1	59.1	73.1	73.8	101.8	129.4	96.1	96.2	60	118.1	901.6	90.1	90.1
70	35.4	30.0	36.4	35.9	45.1	55.6	42.6	43.7	70	59.1	960.7	96.0	96.0
80	14.8	7.7	13.2	13.4	22.3	29.4	23.2	22.6	80	30.8	991.5	99.1	99.1
90	0.0	0.0	0.0	0.0	0.4	8.8	3.8	3.6	90	9.4	1000.9	100.0	100.0

Conical surface Flux(90deg) : 698.1 lm

%lum = 69.7%

%lamp = 69.7%

Conical surface Flux(120deg) : 901.6 lm

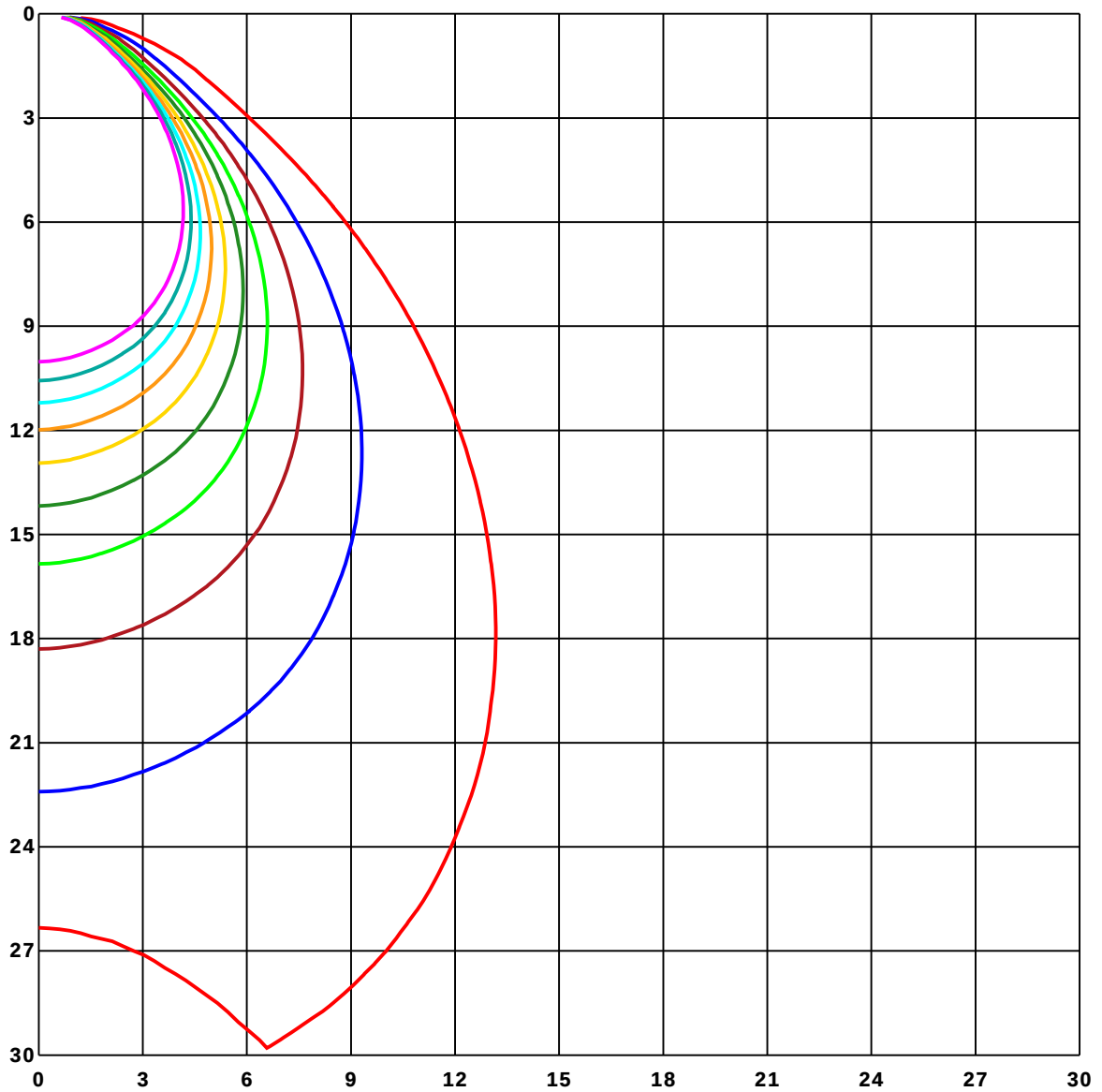
%lum = 90.1%

%lamp = 90.1%

Report No.:

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SPACE ISOLX DIAGRAM



Unit of illumination:lx

5.102
 4.592
 4.082
 3.572
 3.061

2.551
 2.041
 1.531
 1.02
 0.51

C Range: 0 - 360 Deg

C Interval: 45.0 Deg

Test Speed: High

Temperature: 6.25

Operators:

Test Date: 2024/10/15 11:25:44

G Range: 0 - 90 Deg

G Interval: 1.0 Deg

Test System: Huipu CPM-2000B Goniophotometer

Humidity: 1

Test Distance: 6.25 m

Remarks:

Report No.:

Test Time: 2024/10/15 11:25:44

LUMINOUS DISTRIBUTION INTENSITY DATA

Gamma/C	C0	C45	C90	C135	C180	C225	C270	C315	C360	
G0	512.54	506.52	509.57	511.09	512.54	506.52	509.57	511.09	512.54	
G1	512.19	504.61	508.36	509.61	513.36	507.11	508.4	510.74	512.19	
G2	511.56	502.7	507.15	509.34	513.91	508.13	509.14	510.9	511.56	
G3	509.57	501.29	506.05	508.05	513.79	508.09	508.48	509.73	509.57	
G4	509.14	499.22	504.73	507.23	513.71	510.04	508.28	508.95	509.14	
G5	506.6	497.58	503.4	506.13	511.91	507.73	507.5	508.32	506.6	
G6	505.27	495.35	501.09	503.98	511.64	506.95	506.41	506.29	505.27	
G7	501.91	491.45	498.55	501.64	510.0	506.64	505.43	505.12	501.91	
G8	499.73	488.32	496.45	500.39	509.18	505.7	503.87	503.63	499.73	
G9	496.88	485.08	494.41	496.8	506.68	504.45	502.03	501.25	496.88	
G10	494.53	481.41	490.43	495.47	504.57	502.97	499.22	500.0	494.53	
G11	490.66	479.34	487.34	491.02	502.23	501.21	496.48	496.37	490.66	
G12	488.32	474.41	484.88	488.59	499.96	498.87	493.83	493.2	488.32	
G13	484.18	471.68	481.95	484.65	497.42	496.56	491.6	490.2	484.18	
G14	480.51	466.76	477.85	481.56	493.67	494.45	488.09	488.13	480.51	
G15	476.37	463.67	473.95	478.4	490.9	491.64	484.38	485.0	476.37	
G16	473.32	459.41	470.51	474.65	487.58	489.02	481.64	480.7	473.32	
G17	468.71	454.65	466.64	470.78	484.18	485.55	478.09	478.44	468.71	
G18	463.87	449.14	462.19	466.37	480.59	482.11	475.59	474.73	463.87	
G19	458.44	443.55	458.83	461.76	477.19	479.73	470.27	471.68	458.44	
G20	453.4	437.38	453.59	457.85	472.89	475.98	465.39	467.73	453.4	
G21	447.15	431.72	448.44	452.11	468.44	473.13	461.76	463.98	447.15	
G22	440.55	423.75	441.91	446.8	464.38	469.41	456.88	459.45	440.55	
G23	434.53	416.41	435.59	440.74	459.41	466.37	452.77	454.49	434.53	
G24	427.54	408.52	429.57	433.87	454.1	462.19	446.88	449.84	427.54	
G25	419.61	400.23	421.8	427.89	448.52	457.15	440.7	443.91	419.61	
G26	411.37	392.73	414.3	419.84	442.07	453.2	434.3	436.91	411.37	
G27	402.85	383.28	407.07	411.99	435.23	448.09	427.89	430.16	402.85	
G28	394.57	373.52	398.4	404.3	427.77	442.85	420.7	423.44	394.57	
G29	386.09	363.95	390.2	395.66	419.84	436.72	412.97	415.7	386.09	
G30	377.27	354.41	381.13	387.34	411.88	428.59	405.47	408.79	377.27	
G31	367.66	344.69	372.54	379.02	403.91	421.95	397.89	399.8	367.66	
G32	358.2	334.26	363.2	369.26	395.12	413.79	389.06	391.64	358.2	
G33	347.34	323.67	352.85	360.12	386.52	405.82	380.59	382.85	347.34	
G34	337.58	313.59	343.79	350.43	377.7	397.58	370.82	373.79	337.58	
G35	327.03	302.54	333.52	340.63	368.13	389.26	361.41	364.73	327.03	
G36	316.64	292.73	323.71	329.8	359.02	380.16	351.95	356.41	316.64	

C Range: 0 - 360 Deg

C Interval: 45.0 Deg

Test Speed: High

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Operators:

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G Range: 0 - 90 Deg

G Interval: 1.0 Deg

Test System: Huipu CPM-2000B Goniophotometer

Humidity: 1

Test Distance: 6.25 m

Remarks:

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LUMINOUS DISTRIBUTION INTENSITY DATA(Continue 1)

Gamma/C	C0	C45	C90	C135	C180	C225	C270	C315	C360	
G37	305.94	281.09	312.5	319.57	348.87	371.33	341.37	345.0	305.94	
G38	294.92	270.35	302.03	308.4	338.75	361.84	331.52	335.0	294.92	
G39	283.48	259.26	291.13	297.58	328.4	351.99	321.05	324.18	283.48	
G40	272.19	248.16	279.73	287.97	317.66	342.58	309.41	313.75	272.19	
G41	260.59	237.89	269.18	275.43	307.46	331.13	299.02	302.11	260.59	
G42	249.41	226.76	258.01	264.34	296.25	320.31	287.38	290.55	249.41	
G43	237.77	215.82	246.13	252.62	285.31	309.38	275.55	278.87	237.77	
G44	226.25	205.0	235.12	241.13	274.22	298.48	264.26	267.46	226.25	
G45	214.73	194.61	223.44	229.18	263.91	287.85	252.62	255.78	214.73	
G46	203.36	183.91	211.99	217.66	252.5	276.29	241.02	243.75	203.36	
G47	192.19	173.2	200.39	205.74	239.96	265.39	229.34	231.91	192.19	
G48	181.09	162.7	189.22	193.75	229.22	254.22	218.01	220.08	181.09	
G49	170.2	151.95	178.2	181.72	217.3	243.48	206.21	208.36	170.2	
G50	159.73	141.72	166.41	169.92	205.39	232.34	195.2	197.11	159.73	
G51	149.3	131.68	155.2	158.63	193.67	221.37	184.14	185.78	149.3	
G52	139.18	121.72	144.41	147.5	182.5	210.59	173.2	175.12	139.18	
G53	129.3	112.3	133.87	136.33	171.09	200.0	162.62	163.44	129.3	
G54	120.2	102.97	123.83	125.86	160.2	189.45	152.3	153.2	120.2	
G55	110.98	94.1	114.02	115.82	149.38	178.98	142.27	142.42	110.98	
G56	102.27	85.9	104.57	106.02	139.06	168.55	132.42	132.38	102.27	
G57	94.1	78.32	95.94	97.11	128.87	158.44	122.7	122.5	94.1	
G58	86.21	71.21	87.7	88.67	119.73	148.59	113.63	113.2	86.21	
G59	78.98	64.77	80.12	80.86	110.27	138.98	104.65	104.38	78.98	
G60	72.15	59.14	73.13	73.79	101.8	129.38	96.05	96.21	72.15	
G61	65.98	54.06	66.88	67.42	93.28	120.08	88.16	88.2	65.98	
G62	60.55	49.84	61.05	61.72	85.66	111.05	80.78	80.98	60.55	
G63	55.78	45.9	56.25	56.84	78.48	102.23	74.02	74.14	55.78	
G64	51.45	42.54	51.88	52.42	72.07	93.87	67.85	68.01	51.45	
G65	47.73	39.61	48.16	48.71	66.02	85.9	62.38	62.54	47.73	
G66	44.57	37.27	44.77	45.2	60.78	78.52	57.3	57.85	44.57	
G67	41.72	35.59	41.84	42.15	56.25	71.84	52.89	53.71	41.72	
G68	39.1	34.06	39.69	39.53	52.11	65.63	49.06	50.0	39.1	
G69	36.99	32.23	38.09	37.42	48.4	60.35	45.74	46.64	36.99	
G70	35.39	30.0	36.37	35.9	45.08	55.63	42.58	43.67	35.39	
G71	33.67	27.85	34.3	34.1	42.42	51.45	39.8	40.94	33.67	
G72	31.56	25.47	32.07	32.15	40.08	47.7	37.46	38.87	31.56	
G73	29.3	23.09	29.69	29.84	38.44	44.41	35.66	37.19	29.3	

C Range: 0 - 360 Deg

C Interval: 45.0 Deg

Test Speed: High

Temperature: 6.25

Operators:

Test Date: 2024/10/15 11:25:44

G Range: 0 - 90 Deg

G Interval: 1.0 Deg

Test System: Huipu CPM-2000B Goniophotometer

Humidity: 1

Test Distance: 6.25 m

Remarks:

Test Time: 2024/10/15 11:25:44

[illegible]

Remarks: